

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CS-4005-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

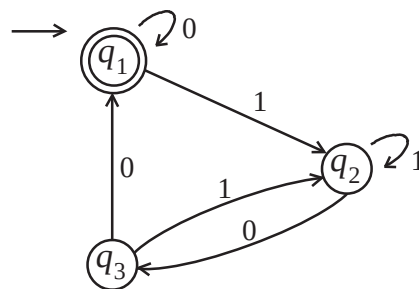
Theory of Computation

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is Mealy Machine? How Finite Automates can be converted into Moore Machine? Explain with the help of example.
b) Design a $(\text{Mod}4)_2$ machine over the alphabet $(0,1)$?
2. Find out the Regular Expression from given DFA



3. Write the CFG for the following language
 - i) $L = \{0^i 1^j 2^k, i = j \text{ or } j = k\}$
 - ii) $L = \{0^n 1^n, n \geq 1\}$
 - iii) $L = \{\text{even palindrome over } (0, 1)\}$

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4. a) Show that the grammar
 $S \rightarrow a / abSb / aAb$
 $A \rightarrow bS / aAAb$ is ambiguous
b) Explain Chomsky Normal Form. Explain with example.
5. Construct the PDA for the following example?
 - i) $a^n b^n \quad n \geq 1$
 - ii) $a^m b^n c^{m+n} \mid m, n \geq 1$
 - iii) $w c w^R \quad w \in (0,1)^*$
6. a) Design a Turing Machine for the language
 $\{L(G) = a^n b^n \mid n \geq 1\}$
b) Describe various types of Turing Machine.
7. a) How P class problem different from NP class problem?
b) What do you mean by Vertex cover problem and Hamiltonian path problem?
8. Write a short notes (any three):
 - i) NP Hard
 - ii) Decidable problem
 - iii) Pumping Lemma for Regular
 - iv) Myhill - Nerode
